



Dore 2023 CS data

A report prepared on behalf of Wye Salmon Association (WSA)

This report takes a very quick look at 2023 WSA CS phosphate data from the Dore sub catchment. The primary objective is to summarise the big picture. Note that all data is presented as RP (mgP/l), rather than raw Hanna readings (ppm as PO₄).

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Upper Dore

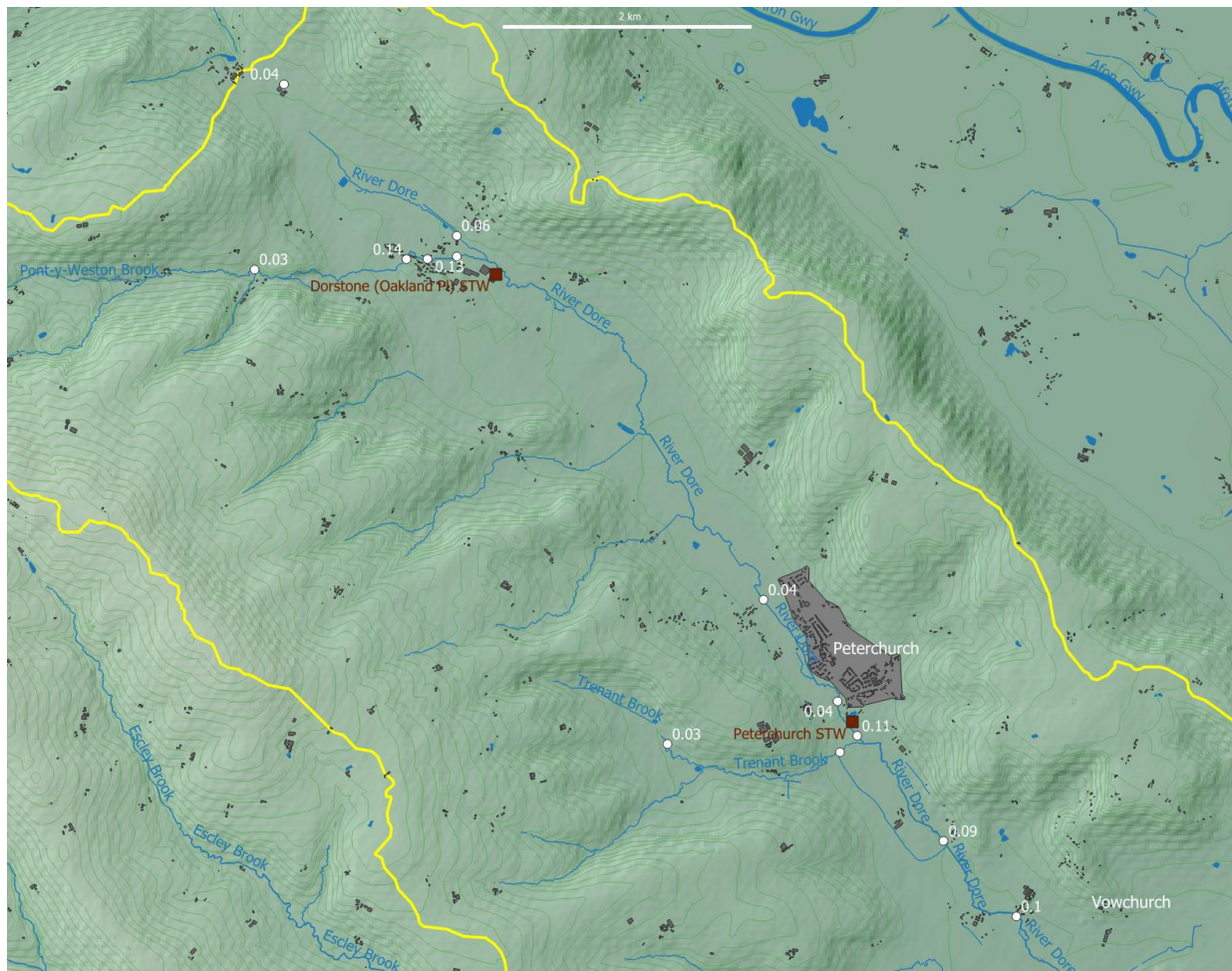


Figure 1: mean RP (Hanna reading / 3.06) at WSA sites within the upper Dore sub catchment, for the period Jan-Sep 2023

Pont y Weston Brook

Pont y Weston Brook rises from unimproved grassland on the western hilltops, before flowing down the valley, mostly through improved pasture en route to Dorstone. Passing through Llanarch Ford (WSA 108), it is clean; 0.03 mgP/l is well below the WFD *good* threshold and not far above *high*. By the time it reached Dorstone (Bridge Cottage, WSA -097), however, RP has increased to 0.14 mgP/l, well into *moderate* classification. It does not increase further as it flows through the village to Church Bridge (WSA140).

Trenant Brook

The Trenant Brook catchment is also heavily dominated by grassland. In its upper reaches it is clean (0.03 mgP/l at WSA 107) like Pont y Weston. The lower monitoring site near Peterchurch shown in figure 1 has been discontinued in 2023. But looking back to 2022, we find the upstream location again at 0.03 mgP/l and the lower location at 0.04 mgP/l. Thus, the Trenant Brook seems to be achieving *good* status throughout its length.

River Dore

The upper reaches of the Dore at The Bage (WSA 074) are also fairly clean at 0.04 mgP/l. But this increases to 0.06 mgP/l just above the Pont y Weston confluence at Dorstone – just falling out of WFD *good* status. The river cleans up again flowing down to Peterchurch. Adjacent to this village, we find two

locations both showing 0.04 mgP/l. After passing Peterchurch STW (input load 856 PE), this has increased to a shocking 0.11 mgP/l – well into *moderate* classification. The admixture of cleaner water from Trenant brook brings this down slightly to 0.09 mg/l at Poston Mill, and this level persists down to Vowchurch. The sampling location much further downstream at Abbey Dore has been discontinued in 2023, but looking back to 2022, we find a mean RP value of 0.04 mgP/l at this location. Thus, the River Dore cleans up below Vowchurch, presumably by a combination of RP ‘decay’ (condensation and metabolism processes) and admixture of cleaner water. At Elm Bridge above Pontrilas we find a 2023 value of 0.05 mgP/l, still achieving WFD *good* status.

It seems that, in the absence of localised hotspots, the Dore has a background level of around 0.04 – 0.05 mgP/l; slightly lower (0.03 mgP/l) in the headwaters.

Worm Brook

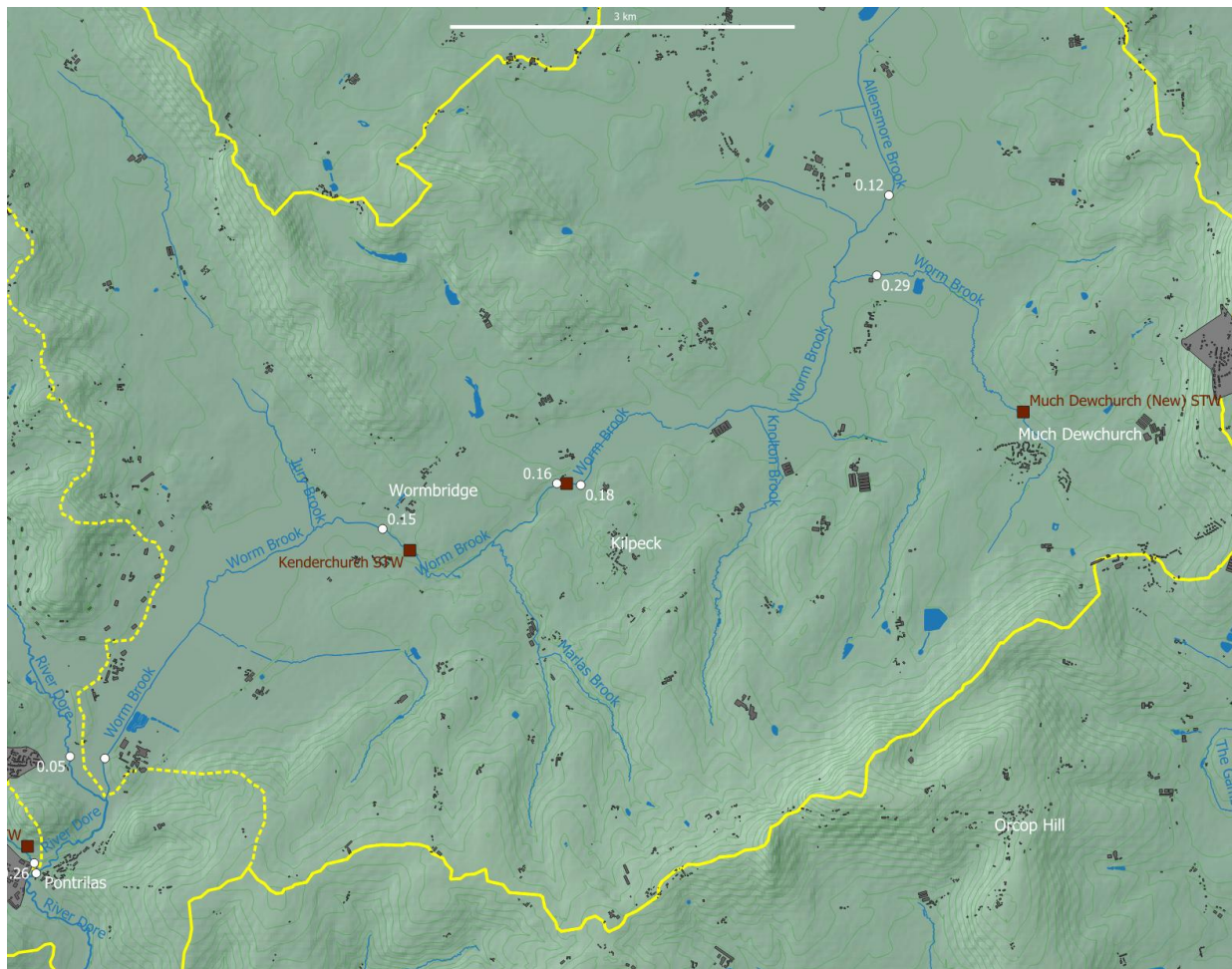


Figure 2: mean RP (Hanna reading / 3.06) at WSA sites within the Worm sub catchment, for the period Jan-Sep 2023

The Worm starts off dirty and stays that way. The upper Worm, below Much Dewchurch is truly dreadful at 0.29 mgP/l. This is WFD *poor* status – not even a *moderate*. Upstream, we have a 120k bird IPU, and Much Dewchurch STW (input load 523 PE). The Allensmore Brook joining from the north-east is a little better at 0.12 mgP/l. Below the confluence we obviously find an intermediate value. Passing Kilpeck and Wormbridge, we see a progressively falling trend, but remaining firmly in the *moderate* WFD classification. The sampling point at Sawmills Footbridge has been discontinued in 2023 but, looking back to 2022 data, we find a value of 0.14 mgP/l at this point – continuing the slight downward trend.

General background levels in the Worm would seem to be in the region of 0.12 – 0.15 mgP/l. The exception is the upper river around Much Dewchurch where we find 2x this level. This is then progressively diluted as the Worm flows downstream. Kenderchurch STW (54 PE) and Kilpeck STW (85 PE) are probably too small to have much impact on the lower river in normal operation. But the larger Much Dewchurch STW is almost certainly having a major impact on the much smaller headwater river. I have estimated that Much Dewchurch STW effluent constitutes about 17% of Q95 flow in the Worm at this point.

Dulas & lower Dore

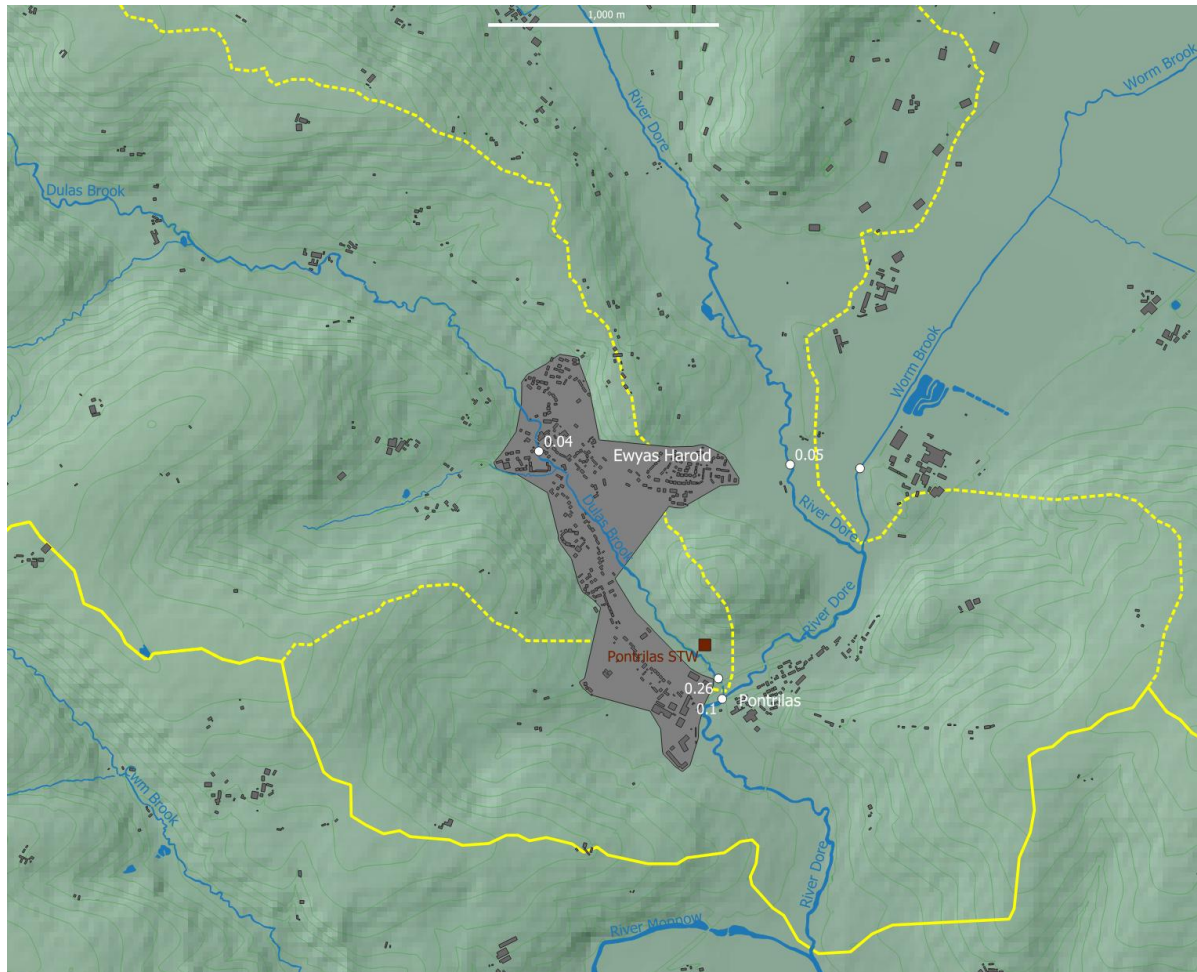


Figure 3: mean RP (Hanna reading / 3.06) at WSA sites within the Dulas and lower Dore sub catchments, for the period Jan-Sep 2023

Dulas Brook

The Dulas catchment is almost exclusively pasture. The brook enters Ewyas Harold in a clean state, similar to the Trenant and upper Dore, at 0.04 mgP/l. This is WFD *good* status. 200m above the Dore confluence, it receives the effluent from Pontillas STW (input load 856 PE), raising the RP concentration to 0.26 mgP/l. I have estimated that effluent from this STW constitutes about 13% of Q95 flow in the Dulas.

Lower Dore

The River Dore below the Dulas confluence simply takes an intermediate value, being a blend of the Worm at 0.14 mgP/l, the Dore at 0.05 mgP/l and the smaller Dulas at 0.26 mgP/l.